

How the players are valued

The method behind the model price on the website, the parameters it uses, and why.

1. What the model price means

The model price is an estimate of the **transfer fee** a player would fetch if he were sold from his current club on a given date. It is not a rating of how good the player is. It answers: given what clubs actually paid in the past for players like this one, what would a club pay today?

The model learns from 15,138 real transfers with a disclosed fee between January 2014 and July 2026. It then applies what it learned to 19,435 active players who are not necessarily for sale.

2. The core idea: start from the Transfermarkt value, then correct it

The model does not predict a fee from scratch. It predicts the **ratio between the fee and the player's Transfermarkt value**:

model price = Transfermarkt value x correction factor

The Transfermarkt value is a good first guess, but it is wrong in systematic ways: on its own it misses the real fee by a median of 50%. The model learns when fees tend to come in above or below it. In the data, players aged 21 or under sold for a median of 1.7 times their Transfermarkt value, and players aged 28 to 31 for 0.6 times. Clubs in England sold at 1.2 times the value, clubs in the United States at 0.5 times. Every parameter below is used to work out this correction factor.

Two reasons for this design:

- **It handles inflation.** Fees rise over time, but so do Transfermarkt values, so the ratio between them stays stable. A model that predicted the fee directly would underprice players in later years.
- **It keeps what Transfermarkt already knows.** Its values already reflect performance, reputation and potential as judged by thousands of users. The model only has to learn what they get wrong.

3. Players by position: what the statistics show

A striker and a goalkeeper are not judged on the same things. Before changing the model, the match statistics of the previous 365 days were compared across positions, on the 5,145 transferred players who played at least 900 minutes that year (317 goalkeeper, 1,671 defender, 1,448 midfield, 1,709 attack).

Average over the previous 365 days	Goalkeeper	Defender	Midfield	Attack
Goals per 90 minutes	0.00	0.05	0.14	0.36
Assists per 90 minutes	0.00	0.07	0.13	0.18
Goals conceded by his team per 90 minutes	1.37	1.36	1.39	1.39
Share of games without a goal conceded	28%	28%	27%	28%
Cards per 90 minutes	0.06	0.22	0.23	0.18
Points per game when he played	1.37	1.44	1.44	1.44
Minutes per game	90	79	72	66
Share of games played in full	99%	74%	53%	38%
Minutes over the year	2588	2242	2247	2074

The positions differ where expected: forwards score about seven times as often as defenders, goalkeepers play every minute, defenders and midfielders collect the most cards. What a team concedes is the same for every position, so on its own it says nothing about position, only about the team.

What matters is how each statistic goes with the fee **within** a position. The table gives the rank correlation with the fee: +1 means the statistic and the fee always rise together, 0 means no link.

Link with the fee, within each position	Goalkeeper	Defender	Midfield	Attack
Goals per 90 minutes	-0.07	+0.09	+0.17	+0.26
Assists per 90 minutes	+0.03	+0.05	+0.15	+0.22
Goals conceded by his team per 90 minutes	-0.28	-0.19	-0.22	-0.19
Share of games without a goal conceded	+0.20	+0.15	+0.17	+0.14
Cards per 90 minutes	-0.09	-0.03	-0.03	-0.08
Points per game when he played	+0.30	+0.28	+0.31	+0.35
Minutes per game	+0.05	+0.09	+0.12	+0.10
Share of games played in full	+0.03	+0.10	+0.13	+0.12
Minutes over the year	+0.41	+0.35	+0.38	+0.36

- **Goals and assists pay for forwards, less for midfielders, not for defenders or goalkeepers.**
- **Goals conceded and clean sheets matter most for goalkeepers.** They also show for the other positions, because they measure the strength of the team as much as the player.
- **Playing time and team results pay for everyone.** Minutes and points per game are the strongest links at every position.

4. How the model adapts to the position

Eleven position statistics were added to the model, which now uses 54 parameters instead of 43. The model knows each player's position and detailed position, so it can give a different weight to each statistic depending on the position: goals count for a forward, goals conceded for a goalkeeper.

Two ways of adapting to position were tested on 5,252 real transfers from 2024-2026, never seen in training. The table gives the typical error (lower is better).

Typical error by position	Transfers	Transfermarkt	Before (43 parameters)	One shared model with position statistics	One separate model per position
All	5,252	0.942	0.761	0.758	0.796
Goalkeeper	327	1.019	0.828	0.831	0.923
Defender	1,514	0.990	0.780	0.780	0.820
Midfield	1,435	0.938	0.758	0.754	0.792
Attack	1,976	0.893	0.737	0.732	0.757

- **One separate model per position is worse** (0.796 against 0.761). Each model learns from far fewer transfers, 959 for goalkeepers, and loses what the positions have in common: age, club, country and Transfermarkt value work the same way for everyone.

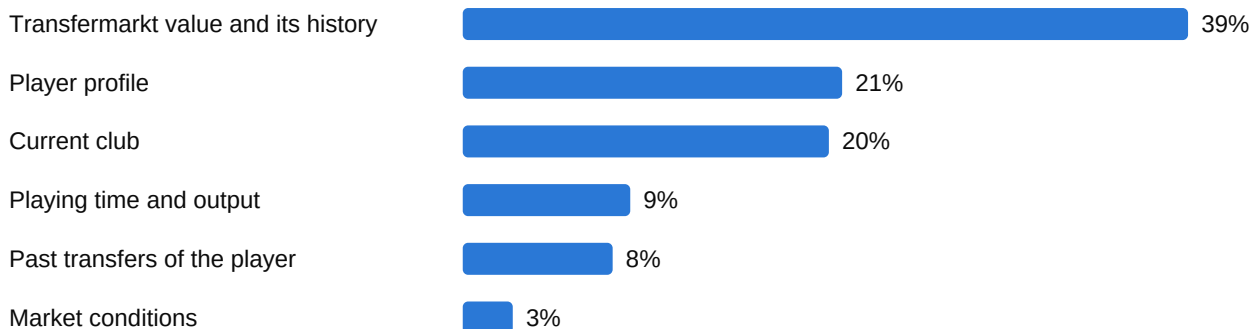
- **One shared model with position statistics is the one kept.** The error goes from 0.761 to 0.758. The gain is small and comes from midfielders and forwards; goalkeepers and defenders do not improve.
- **Why the gain is small.** The Transfermarkt value already reflects performance, match data exists for only 14 top divisions (about half of the transfers), and the statistics available for defenders and goalkeepers are team results, not individual actions.
- **Not available: challenges won, tackles, saves, expected goals prevented.** The data source records only goals, assists, cards and minutes for each game. Goals conceded, clean sheets and team results while the player was on the pitch stand in for them. Individual defensive and goalkeeping statistics would need a second data source.

Position statistics that weigh most for each position (share of the correction):

Position	Heaviest position statistics
Goalkeeper	minutes per game (1.8%), share of games played in full (1.0%), points per game won by his team when he played (0.7%)
Defender	minutes per game (1.2%), points per game won by his team when he played (0.7%), share of games played in full (0.7%)
Midfield	minutes per game (1.3%), goals plus assists per 90 minutes (0.8%), points per game won by his team when he played (0.8%)
Attack	goals plus assists per 90 minutes (1.3%), minutes per game (1.2%), share of games played in full (0.7%)

5. The parameters and why each is there

The website model uses 54 parameters in six groups. The chart shows how much each group contributes to the correction factor (share of the model's total gain).



Group	Parameters	Why they are used
Transfermarkt value and its history	Last value at least 30 days old; change over 6 and 12 months; distance from the player's peak value; number of valuations so far; years since the first valuation; age of the last valuation.	The value is the starting point. Its trend shows whether the player is on the way up or down. The number of valuations measures how closely the player is followed, and little-followed players are valued less reliably.
Player profile	Age; position and detailed position; nationality; height; preferred foot.	Age is the strongest single driver after the value: clubs pay for the years a player has left and for resale potential. Fees also differ by position and nationality for the same value.

Group	Parameters	Why they are used
Current club	Country and league; points per game last season; squad value; average squad value in its league; transfer income and number of sales over 3 years; how its past sales compared with Transfermarkt values; whether it is a youth or reserve team.	Who sells matters: the same Transfermarkt value does not fetch the same fee from every club or country. The club's past sales show whether it tends to sell above or below value. The country is the most important parameter in this group.
Past transfers of the player	Time at the current club; fee the current club paid; last and highest fee ever paid for him; number of previous transfers.	The fee a club paid is a reference point when it sells. Time at the club is also the only available stand-in for contract length: a player who arrived recently usually has a long contract left.
Playing time and output	Minutes, games, goals and assists over the previous 365 days; goals plus assists per 90 minutes; minutes in league and in European competitions; minutes for the current club; change versus the year before; career totals. New, for the positions: goals and assists per 90 minutes separately; goals conceded, goals scored and goal difference of his team per 90 minutes he played; share of games without a goal conceded; points per game when he played; cards per 90 minutes; minutes per game; share of games played in full; career assists.	Clubs pay for players who play and produce, and what counts as producing depends on the position. These add little (9%) because the Transfermarkt value already reflects performance, and because match data exists for only 14 top divisions.
Market conditions	How fees compared with Transfermarkt values across the whole market over the previous year; summer or winter window.	Captures periods when the whole market pays more or less than Transfermarkt values.

6. The weight of each parameter

Share of the model's total gain carried by each parameter: how much it improves the fit while the model learns. The column Before is the model with 43 parameters, whose six group percentages were given earlier; Now is the model with position statistics. Each column adds up to 100%.

Parameter	Group	Before (43)	Now (54)
Last Transfermarkt value (at least 30 days old)	Transfermarkt value and its history	14.7%	14.4%
Number of Transfermarkt valuations so far	Transfermarkt value and its history	13.6%	12.1%
Age	Player profile	11.3%	11.8%
Country of the club	Current club	8.5%	8.4%
Nationality	Player profile	5.5%	5.2%
Years since the first valuation	Transfermarkt value and its history	5.7%	4.9%
Club's transfer income over 3 years	Current club	4.8%	4.8%
Time at the current club	Past transfers of the player	3.9%	3.6%
Club's past sales compared with Transfermarkt values	Current club	3.4%	3.5%

Parameter	Group	Before (43)	Now (54)
Change in value over 12 months	Transfermarkt value and its history	2.1%	2.6%
Age of the last valuation, in days	Transfermarkt value and its history	2.2%	2.2%
Fees compared with Transfermarkt values across the market, last year	Market conditions	2.4%	2.2%
Change in value over 6 months	Transfermarkt value and its history	2.1%	1.8%
Detailed position (e.g. left winger)	Player profile	1.9%	1.8%
Height	Player profile	1.7%	1.6%
Last fee paid for the player	Past transfers of the player	1.7%	1.6%
Highest fee ever paid for the player	Past transfers of the player	1.2%	1.5%
Value compared with the player's peak value	Transfermarkt value and its history	1.2%	1.3%
Fee the current club paid	Past transfers of the player	1.4%	1.0%
Club's points per game last season	Current club	1.3%	1.0%
Minutes for the current club, last 365 days	Playing time and output	1.3%	1.0%
Change in minutes versus the year before	Playing time and output	1.1%	0.9%
Squad value of the club	Current club	0.9%	0.9%
Minutes per game	Playing time and output	new	0.8%
Career minutes	Playing time and output	0.6%	0.7%
Average squad value in the club's league	Current club	0.6%	0.7%
Goal difference of his team per 90 minutes he played	Playing time and output	new	0.6%
Share of games played in full	Playing time and output	new	0.6%
Summer or winter window	Market conditions	0.6%	0.5%
Number of previous transfers	Past transfers of the player	0.6%	0.5%
Goals conceded by his team per 90 minutes he played	Playing time and output	new	0.5%
Points per game won by his team when he played	Playing time and output	new	0.5%
Share of his games without a goal conceded	Playing time and output	new	0.4%
Goals plus assists per 90 minutes	Playing time and output	0.5%	0.4%
Assists per 90 minutes	Playing time and output	new	0.4%
League of the club	Current club	0.4%	0.4%
Goals scored by his team per 90 minutes he played	Playing time and output	new	0.4%
Cards per 90 minutes (a red counts as three yellows)	Playing time and output	new	0.3%
Career games	Playing time and output	0.5%	0.3%

Parameter	Group	Before (43)	Now (54)
Number of sales by the club over 3 years	Current club	0.4%	0.3%
Goals per 90 minutes	Playing time and output	new	0.2%
Minutes in the year before	Playing time and output	0.3%	0.2%
Minutes in the league, last 365 days	Playing time and output	0.3%	0.2%
Position (goalkeeper, defender, midfield, attack)	Player profile	0.3%	0.2%
Minutes, last 365 days	Playing time and output	0.3%	0.2%
Career goals	Playing time and output	0.2%	0.1%
Games, last 365 days	Playing time and output	0.2%	0.1%
Youth or reserve team	Current club	0.1%	0.1%
Goals, last 365 days	Playing time and output	0.1%	0.1%
Preferred foot	Player profile	0.1%	0.1%
Career assists	Playing time and output	new	0.1%
Assists, last 365 days	Playing time and output	0.1%	0.0%
Minutes in European competitions, last 365 days	Playing time and output	0.1%	0.0%
Match data available for the player	Playing time and output	0.0%	0.0%

7. The weight of each parameter for each position

The same 54 parameters, for the players of each position. The measure is the share of the correction factor that comes from the parameter, averaged over the test transfers of that position (mean absolute SHAP value). It is not the same measure as in section 6, so the column All differs from it: a parameter can improve the fit a lot while moving each price a little. Each column adds up to 100%; group totals are in bold.

Parameter	All	Goalkeeper	Defender	Midfield	Attack
Transfermarkt value and its history	26.5%	26.7%	26.8%	26.3%	26.3%
Last Transfermarkt value (at least 30 days old)	11.8%	11.0%	11.8%	11.5%	12.0%
Number of Transfermarkt valuations so far	6.4%	6.3%	6.6%	6.5%	6.3%
Change in value over 6 months	2.3%	2.3%	2.4%	2.4%	2.3%
Years since the first valuation	1.8%	2.6%	1.8%	1.8%	1.7%
Change in value over 12 months	1.4%	1.6%	1.5%	1.5%	1.3%
Age of the last valuation, in days	1.3%	1.6%	1.4%	1.3%	1.3%
Value compared with the player's peak value	1.3%	1.4%	1.3%	1.4%	1.3%
Player profile	22.2%	23.1%	21.9%	21.6%	22.7%
Age	10.4%	11.0%	10.5%	10.4%	10.2%
Nationality	6.0%	6.3%	6.1%	6.0%	5.8%
Height	3.0%	4.4%	3.2%	2.7%	2.9%

Parameter	All	Goalkeeper	Defender	Midfield	Attack
Detailed position (e.g. left winger)	2.4%	1.0%	1.8%	2.2%	3.3%
Position (goalkeeper, defender, midfield, attack)	0.3%	0.3%	0.2%	0.2%	0.4%
Preferred foot	0.1%	0.1%	0.2%	0.1%	0.1%
Current club	18.3%	18.7%	18.7%	18.6%	17.7%
Country of the club	8.1%	8.8%	8.2%	8.2%	8.0%
Club's transfer income over 3 years	4.0%	3.7%	4.2%	4.0%	3.9%
Club's past sales compared with Transfermarkt values	1.7%	1.6%	1.7%	1.7%	1.7%
Club's points per game last season	1.5%	1.6%	1.6%	1.6%	1.5%
League of the club	1.0%	1.1%	1.1%	1.1%	0.9%
Average squad value in the club's league	0.7%	0.7%	0.7%	0.7%	0.6%
Squad value of the club	0.5%	0.7%	0.5%	0.5%	0.5%
Number of sales by the club over 3 years	0.4%	0.4%	0.4%	0.5%	0.4%
Youth or reserve team	0.2%	0.2%	0.2%	0.2%	0.2%
Playing time and output	14.8%	15.0%	14.6%	15.0%	14.8%
Minutes for the current club, last 365 days	2.8%	2.8%	2.9%	2.8%	2.6%
Career minutes	1.7%	1.8%	1.8%	1.7%	1.6%
Minutes per game	1.3%	1.8%	1.2%	1.3%	1.2%
Change in minutes versus the year before	1.1%	0.9%	1.0%	1.2%	1.1%
Goals plus assists per 90 minutes	0.9%	0.7%	0.6%	0.8%	1.3%
Share of games played in full	0.7%	1.0%	0.7%	0.7%	0.7%
Points per game won by his team when he played	0.7%	0.7%	0.7%	0.8%	0.7%
Goals conceded by his team per 90 minutes he played	0.6%	0.6%	0.7%	0.6%	0.6%
Goal difference of his team per 90 minutes he played	0.6%	0.6%	0.6%	0.6%	0.6%
Career games	0.6%	0.6%	0.6%	0.6%	0.6%
Cards per 90 minutes (a red counts as three yellows)	0.5%	0.4%	0.5%	0.5%	0.4%
Goals per 90 minutes	0.5%	0.5%	0.4%	0.4%	0.5%
Assists per 90 minutes	0.4%	0.5%	0.4%	0.4%	0.4%
Goals scored by his team per 90 minutes he played	0.4%	0.4%	0.4%	0.4%	0.4%
Share of his games without a goal conceded	0.3%	0.3%	0.4%	0.3%	0.3%
Games, last 365 days	0.3%	0.3%	0.3%	0.3%	0.3%
Minutes in the league, last 365 days	0.2%	0.3%	0.3%	0.2%	0.2%
Goals, last 365 days	0.2%	0.2%	0.2%	0.2%	0.3%
Minutes in the year before	0.2%	0.2%	0.2%	0.2%	0.2%

Parameter	All	Goalkeeper	Defender	Midfield	Attack
Minutes, last 365 days	0.2%	0.2%	0.2%	0.2%	0.2%
Career goals	0.2%	0.1%	0.2%	0.2%	0.2%
Career assists	0.1%	0.1%	0.1%	0.1%	0.1%
Minutes in European competitions, last 365 days	0.1%	0.1%	0.1%	0.1%	0.1%
Assists, last 365 days	0.0%	0.0%	0.0%	0.1%	0.1%
Match data available for the player	0.0%	0.0%	0.0%	0.0%	0.0%
Past transfers of the player	11.5%	10.3%	11.4%	11.4%	12.0%
Time at the current club	3.2%	3.1%	3.4%	3.1%	3.2%
Highest fee ever paid for the player	3.1%	2.8%	3.0%	3.2%	3.3%
Last fee paid for the player	2.6%	2.1%	2.4%	2.5%	2.8%
Fee the current club paid	2.1%	1.8%	2.0%	2.1%	2.2%
Number of previous transfers	0.5%	0.5%	0.5%	0.5%	0.5%
Market conditions	6.7%	6.1%	6.7%	7.0%	6.5%
Fees compared with Transfermarkt values across the market, last year	4.3%	3.9%	4.3%	4.4%	4.2%
Summer or winter window	2.4%	2.3%	2.4%	2.5%	2.3%

The weights are close across positions because most of the correction comes from parameters that work the same way for everyone. The differences are where expected: height and the minutes per game weigh more for goalkeepers, goals plus assists and the detailed position more for forwards.

8. What is deliberately left out

Left out	Why
Anything about the buyer (its league, wealth, past spending)	A player who is not being sold has no buyer. The version of the model that does know the buyer is more accurate (38% median error against 41%) and is the one used to judge real transfers in the thesis.
Anything dated after the valuation date	Transfermarkt often updates a value after a transfer, using the fee. Using it would let the model see the answer. Hence the 30-day rule on values, and performance counted only before the date.
Current international caps, current club, current value	The source only gives today's figures, not those at the time of each past transfer.
Contract length	Not available historically: Transfermarkt only shows current contracts. This is the most important missing parameter.
Challenges won, tackles, saves, expected goals	Not in the data source. Team results while the player was on the pitch are used instead (section 4).

9. How the model learns

- **Algorithm.** Gradient boosting (LightGBM): about a thousand small decision trees, each correcting the errors of the previous ones. It finds on its own effects that are not straight lines, such as age, and combinations such as goals together with position.

- **One model per year, trained on the past only.** The price on 1 July 2023 comes from a model trained only on transfers before that date, and so on for each year from 2022 to 2026. A past price is what the model would have said at the time.
- **Sample rules.** Permanent transfers with a disclosed fee of at least EUR 50,000. Free transfers, undisclosed fees and suspected loan fees are excluded because they give no usable price.
- **Who is valued.** Every player given a Transfermarkt value in the 18 months before the date.
- **Scale-down for expensive players.** The raw correction is too strong when the Transfermarkt value is high: bonuses for being young, being sold from England or having cost a lot are learned mostly on cheap players, and for a star the Transfermarkt value already contains them. On unseen transfers above 40m, only about 60% of the predicted correction showed up in the real fee. The correction is therefore left untouched below EUR 5m, multiplied by 0.65 above EUR 40m, and in between in proportion. These numbers are estimated on past transfers with out-of-sample predictions, again for each yearly model. Example: Estêvão (Transfermarkt value EUR 80m) came out at EUR 166m before this step and at EUR 126m after.

10. How far to trust it

Check on 5,252 real transfers from 2024-2026, never seen in training	Median error	R2 (log)
Transfermarkt value alone	50%	0.64
Website model before position statistics (43 parameters)	41%	0.76
Website model (no buyer information, 54 parameters)	41%	0.77
Full model (knows the buyer)	38%	0.82

- **The model range** on the website is an 80% interval: in the year before each date, four real fees in five fell inside it. Its width depends on the Transfermarkt value: the price divided and multiplied by 3.0 for players under 1m, 2.4 from 1m to 5m, 2.0 from 5m to 20m and 1.4 above 20m. This is wide. The model is reliable on averages over many players, much less on one player.
- **Selection.** The model only learns from players who were sold. Players whom clubs refuse to sell may differ from them in ways the data does not show.
- **The most expensive players.** Above 40m the model is only slightly more accurate than the Transfermarkt value alone; its price there should be read as Transfermarkt with a small correction.
- **Lower leagues.** Errors are largest for cheap players and for clubs outside the 14 covered leagues.
- **Full statistics** on the reliability of the prices are in Statistics.pdf.

Part 2. The full model: the price for a given buyer

The model that also knows who is buying. Used on the website when a buying club is chosen.

11. Two models, two questions

	Website model (Part 1)	Full model (Part 2)
Question it answers	What fee would this player fetch if he were sold, whoever the buyer?	What fee would this player fetch if this particular club bought him?
Parameters	54: the player, his current club, the market	68: the same 54, plus 14 about the buyer and the deal
Median error on 5,252 unseen transfers	41%	38%
Typical error (log scale)	0.758	0.673
R2 (log)	0.77	0.82
Where it is used	Website, when no buyer is chosen; the five-year history of each player	Website, when a buying club is chosen; judging real transfers in the thesis

The full model is more accurate because the buyer explains a large part of the fee: the same player costs more when a rich club buys him. It cannot be the default on the website, because a player who is not being sold has no buyer. It works as soon as a buyer is named, which is what the buying club option does.

Apart from the extra parameters, the full model is built exactly like the website model: same transfers, same target (fee divided by Transfermarkt value), same algorithm, same position statistics.

12. The 14 parameters about the buyer and the deal

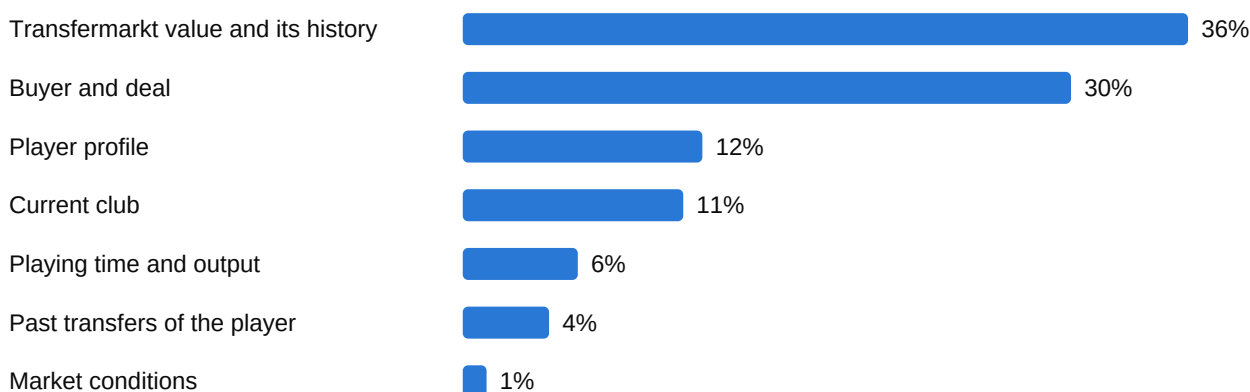
Together they carry 30% of the full model's total gain.

Parameter	Weight	Why it matters
Buying club's transfer spending over 3 years	9.3%	A club that spends a lot pays more for the same player.
Country of the buying club	7.5%	Clubs in England pay more than clubs elsewhere for the same player.
Buying club's past purchases compared with Transfermarkt values	6.3%	Some clubs systematically pay above or below value.
Squad value of the buying club	2.2%	A measure of the buyer's sporting and financial level.
Year of the transfer	1.3%	Lets the model follow the market over time.
Buying club's points per game last season	0.9%	Stronger clubs buy at higher prices.
Average squad value in the buyer's league	0.8%	The wealth of the league the player moves to.
League of the buying club	0.6%	League-specific price levels beyond the country.
Number of purchases by the buying club over 3 years	0.5%	How active the buyer is on the market.
Buyer and seller in the same country	0.4%	Domestic deals are priced differently from moves abroad.

Parameter	Weight	Why it matters
Buyer is a youth or reserve team	0.2%	Signings for a second team are cheaper.
Deal in the last week of the window	0.2%	Last-minute deals are made under time pressure.
Player loaned back to the seller after the sale	0.1%	The buyer pays for the future, not for immediate use.
Buyer and seller in the same league	0.0%	Deals inside one league could be priced differently; the model finds almost no effect.

13. The weight of each parameter in the full model

Share of the full model's total gain by group, then for each of the 68 parameters. The last column gives the weight of the same parameter in the website model (section 6) for comparison. Buyer and deal parameters are in bold. Each column adds up to 100%.



Parameter	Group	Full model	Website model
Last Transfermarkt value (at least 30 days old)	Transfermarkt value and its history	15.3%	14.4%
Number of Transfermarkt valuations so far	Transfermarkt value and its history	12.3%	12.1%
Buying club's transfer spending over 3 years	Buyer and deal	9.3%	not used
Country of the buying club	Buyer and deal	7.5%	not used
Buying club's past purchases compared with Transfermarkt values	Buyer and deal	6.3%	not used
Age	Player profile	6.0%	11.8%
Country of the club	Current club	3.9%	8.4%
Years since the first valuation	Transfermarkt value and its history	3.7%	4.9%
Nationality	Player profile	3.0%	5.2%
Club's transfer income over 3 years	Current club	2.5%	4.8%
Squad value of the buying club	Buyer and deal	2.2%	not used
Club's past sales compared with Transfermarkt values	Current club	2.2%	3.5%

Parameter	Group	Full model	Website model
Time at the current club	Past transfers of the player	1.9%	3.6%
Detailed position (e.g. left winger)	Player profile	1.7%	1.8%
Change in value over 12 months	Transfermarkt value and its history	1.4%	2.6%
Age of the last valuation, in days	Transfermarkt value and its history	1.3%	2.2%
Change in value over 6 months	Transfermarkt value and its history	1.3%	1.8%
Year of the transfer	Buyer and deal	1.3%	not used
Last fee paid for the player	Past transfers of the player	1.0%	1.6%
Buying club's points per game last season	Buyer and deal	0.9%	not used
Height	Player profile	0.9%	1.6%
Fees compared with Transfermarkt values across the market, last year	Market conditions	0.8%	2.2%
Average squad value in the buyer's league	Buyer and deal	0.8%	not used
Club's points per game last season	Current club	0.6%	1.0%
Value compared with the player's peak value	Transfermarkt value and its history	0.6%	1.3%
Career minutes	Playing time and output	0.6%	0.7%
Squad value of the club	Current club	0.6%	0.9%
League of the buying club	Buyer and deal	0.6%	not used
Highest fee ever paid for the player	Past transfers of the player	0.6%	1.5%
Number of purchases by the buying club over 3 years	Buyer and deal	0.5%	not used
Change in minutes versus the year before	Playing time and output	0.5%	0.9%
Fee the current club paid	Past transfers of the player	0.4%	1.0%
Goal difference of his team per 90 minutes he played	Playing time and output	0.4%	0.6%
Career games	Playing time and output	0.4%	0.3%
Number of sales by the club over 3 years	Current club	0.4%	0.3%
Minutes per game	Playing time and output	0.4%	0.8%
Summer or winter window	Market conditions	0.4%	0.5%
Number of previous transfers	Past transfers of the player	0.4%	0.5%
Minutes for the current club, last 365 days	Playing time and output	0.4%	1.0%
Buyer and seller in the same country	Buyer and deal	0.4%	not used
Share of games played in full	Playing time and output	0.3%	0.6%

Parameter	Group	Full model	Website model
Points per game won by his team when he played	Playing time and output	0.3%	0.5%
Cards per 90 minutes (a red counts as three yellows)	Playing time and output	0.3%	0.3%
League of the club	Current club	0.3%	0.4%
Average squad value in the club's league	Current club	0.3%	0.7%
Position (goalkeeper, defender, midfield, attack)	Player profile	0.3%	0.2%
Goals conceded by his team per 90 minutes he played	Playing time and output	0.3%	0.5%
Goals plus assists per 90 minutes	Playing time and output	0.3%	0.4%
Goals scored by his team per 90 minutes he played	Playing time and output	0.2%	0.4%
Goals per 90 minutes	Playing time and output	0.2%	0.2%
Share of his games without a goal conceded	Playing time and output	0.2%	0.4%
Assists per 90 minutes	Playing time and output	0.2%	0.4%
Minutes in the year before	Playing time and output	0.2%	0.2%
Buyer is a youth or reserve team	Buyer and deal	0.2%	not used
Minutes in the league, last 365 days	Playing time and output	0.2%	0.2%
Deal in the last week of the window	Buyer and deal	0.2%	not used
Youth or reserve team	Current club	0.1%	0.1%
Minutes, last 365 days	Playing time and output	0.1%	0.2%
Career goals	Playing time and output	0.1%	0.1%
Preferred foot	Player profile	0.1%	0.1%
Career assists	Playing time and output	0.1%	0.1%
Player loaned back to the seller after the sale	Buyer and deal	0.1%	not used
Games, last 365 days	Playing time and output	0.1%	0.1%
Goals, last 365 days	Playing time and output	0.0%	0.1%
Assists, last 365 days	Playing time and output	0.0%	0.0%
Buyer and seller in the same league	Buyer and deal	0.0%	not used
Minutes in European competitions, last 365 days	Playing time and output	0.0%	0.0%
Match data available for the player	Playing time and output	0.0%	0.0%

With the buyer known, the weight of the player's own profile and of his current club falls: part of what age, nationality or the selling country explained in the website model was in fact a guess about the kind of club that would buy him.

14. The buying club option on the website

- **What it shows.** For the chosen club, the fee the full model expects if that club bought each player on 1 July 2026. The table, the difference with Transfermarkt and the range all switch to that price.

- **Which clubs.** The 508 clubs of the 30 leagues covered by the data in the 2025/26 season. The buyer's figures (spending, squad value, points per game, past purchases) are those known on 1 July 2026.
- **How it is computed.** The full model is trained on the 14,989 transfers before 1 July 2026, then run once for each club on all 19,435 players: 9,872,980 prices in total. The deal is assumed to be a normal summer transfer, not in the last week of the window and without a loan back.
- **How much the buyer changes the price.** For the typical player, the price with a buyer in the top tenth of clubs is 3.8 times the price with a buyer in the bottom tenth.
- **The range.** An 80% interval whose width depends on the Transfermarkt value: the price divided and multiplied by 2.6 under 1m, 2.1 from 1m to 5m, 1.9 from 5m to 20m and 1.5 above 20m. The same scale-down for expensive players applies (correction multiplied by 0.72 above EUR 80m).

Example: Erling Haaland. With no buyer chosen the website model gives EUR 220m. With a buyer:

Buying club	League	Price
NEOM SC	Saudi Pro League	EUR 270m
Al-Qadsiah	Saudi Pro League	EUR 265m
Al-Nassr	Saudi Pro League	EUR 259m
Al-Ahli	Saudi Pro League	EUR 259m
Al-Ittihad	Saudi Pro League	EUR 259m
...		
Livingston FC	Scottish Premiership	EUR 112m
UTA Arad	Superliga (Romania)	EUR 108m
Asteras Tripoli	Super League 1	EUR 105m
FC Arges	Superliga (Romania)	EUR 105m
NK Varazdin	Supersport Hnl	EUR 103m

Limits specific to this option:

- **Unlikely pairs are extrapolations.** The model learned from transfers that happened, where rich clubs bought expensive players and small clubs cheap ones. The price of a low-value player for a top club, or of a star for a small club, is far from anything the model has seen and should not be trusted. The website marks a price with * when the player's Transfermarkt value is below the cheapest 5% or above the most expensive of the players bought by clubs of that league over the last five years.
- **It is a price, not a probability.** The option says what the fee would be if the deal happened, not whether the club could afford it or the player would go.
- **The seller is not modelled as negotiating.** A club that does not want to sell asks for more than the model price, whoever the buyer.
- **Only the latest date.** Buyer prices are computed for 1 July 2026; the five-year history shown for each player stays the model without a buyer.